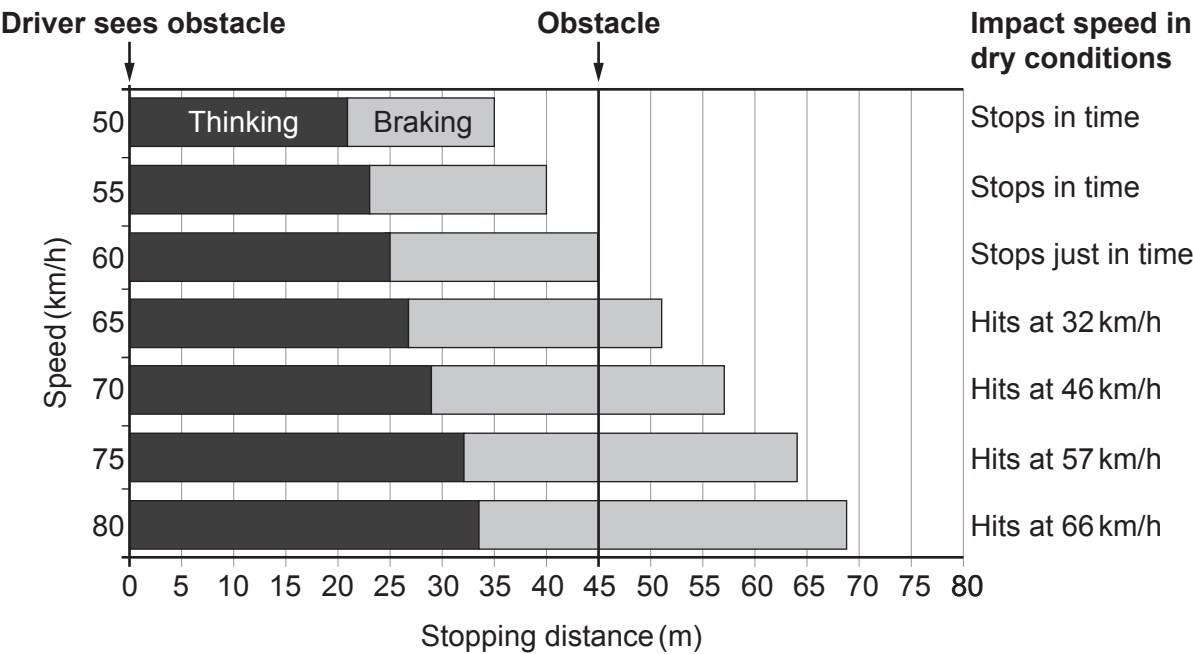


7. The chart below is used by traffic collision investigators. It gives the thinking, braking and stopping distances of cars driven at different speeds by an alert driver on a dry road.

An alert driver notices an obstacle 45 m away on the road ahead. The position of this obstacle is represented by the dark vertical line. If there is a collision, the chart also shows the impact speed with the obstacle.



- (a) State how the following information in the chart for a speed of 70 km/h would compare if the tyre treads on the car are worn below the legal limit. [3]

(i) Thinking distance

.....

(ii) Braking distance

.....

(iii) Impact speed

.....



- (b) Use the information opposite to answer the following questions about a car travelling at 60 km/h which **decelerates to a stop**.
(10 km/h = 2.8 m/s)

(i) **Complete** the following table. [4]

Initial speed (km/h)	Initial speed (m/s)	Thinking distance (m)	Braking distance (m)	Stopping distance (m)
60				

(ii) Using the information in the table, calculate the **thinking time** of the driver. [3]

thinking time = s

(iii) Use the equation:

$$v^2 = u^2 + 2ax$$

and the information in the table to calculate the deceleration of the car. [3]

deceleration = m/s²

- (c) The chart shows that a car travelling faster than 60 km/h will not stop in time and therefore collides with the obstacle.
- (i) Explain, in terms of Newton's 1st Law, how seat belts provide protection during a collision. [2]

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only

(ii) Explain, in terms of Newton's 3rd law, why the car and the obstacle are damaged during a collision. [2]

.....

.....

.....

(iii) Explain, in terms of Newton's 2nd law, why cars have crumple zones. [2]

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(d) The Royal Society for the Prevention of Accidents (RoSPA) says that “The majority of pedestrian casualties (in the UK) occur in built-up areas: 22 of the 26 child pedestrians and 264 of the 372 adult pedestrians who were killed in 2017, died on roads in built-up areas.”
They strongly feel that reducing the speed limit in built-up areas from 50 km/h to 35 km/h would greatly improve these accident statistics.
Explain whether you agree with RoSPA. [3]

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END OF PAPER

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